VARIABLE INFILTRATION CAPACITY (VIC) MODEL

ERT 474/574

Open-Source Hydro Data Analytics

Oct 28th 2026

University at Buffalo The State University of New York



Announcement

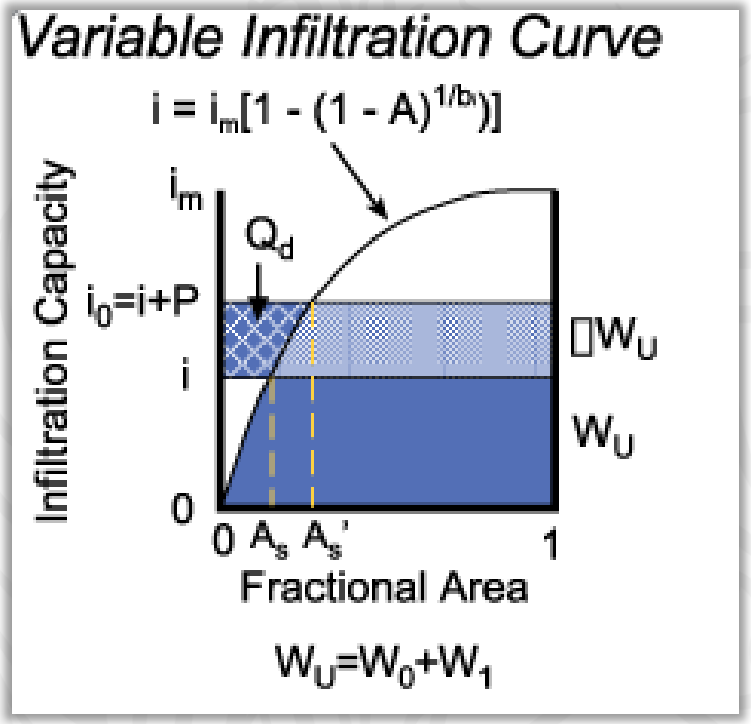
- Homework #6
  - Run the VIC model for FindleyLake Example

An Overview of the VIC Model

- Hydrology model comparison/selection
- VIC model features
  - Cell size
  - Sub-grid representations
- VIC model processes
  - Vegetation
  - Snow
  - Evapotranspiration
  - Runoff/Infiltration
  - Baseflow

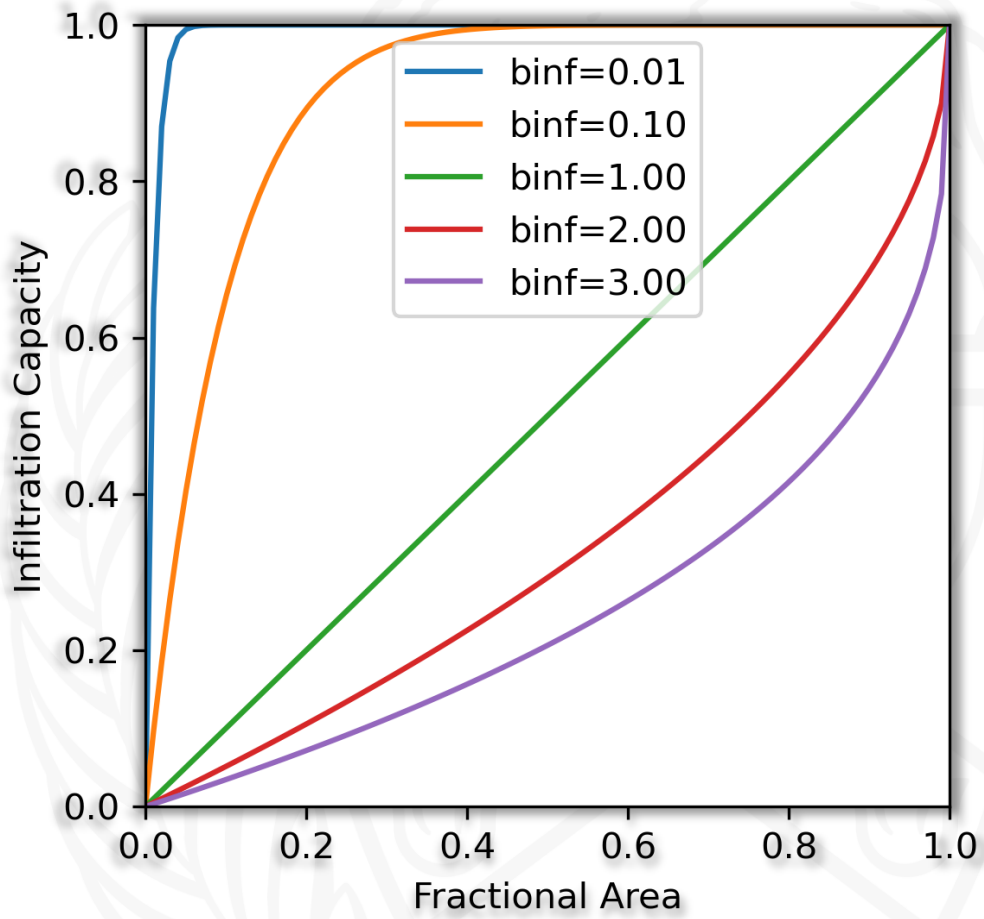
Runoff-Soil Infiltration

- Surface runoff/soil infiltration defined by the **variable infiltration curve** [Wood et al., 1992]
- Scales maximum infiltration with **a non-linear function** of fractional saturated area
  - Enables runoff calculations **for subgrid-scale areas**
- Curve shape defined by b_{inf} parameter (typically between 10^{-5} and 0.4)
  - Amount of infiltration capacity relative to the saturated gridcell area
- Greater value of b_{inf} yields lower infiltration and more runoff (Q_d)



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CONCEPT

Important concept about hydrologic modeling



Important concept

- Overall workflow
- Configuration file
- Initial conditions / State files
- Model Spin-up
- *Flux* versus *State* variables

Overall workflow




```
git clone
```

```
make
```

vic.exe

- Domain file
- Parameter file
- Meteorological forcing
- Initial condition file*

Example config file:

global_param.Stehekin.L2015.txt

vic.exe -g [config file]

Configuration files

- The configuration file is the backbone of any hydrologic model setup. It defines how the model runs, including:
- **Model parameters**: Time step, simulation period, solver settings.
- **Input/output paths**: Locations of input data (e.g., precipitation, temperature) and where outputs are saved.
- **Modules and processes**: Which hydrologic processes are activated (e.g., infiltration, evapotranspiration, routing).
- **Spatial and temporal resolution**: Grid size, time step intervals.

Example: VIC configuration file

```
DOMAIN          /workspaces/VIC_sample_data/image/Stehekin/parameters/domain.Stehekin.0.0625_deg.nc
```

```
FORCING1      /workspaces/VIC_sample_data/image/Stehekin/forcings/Stehekin_image_test.forcings_10days.0.0625_deg.
FORCE_TYPE    AIR_TEMP      tas
FORCE_TYPE    PREC          prcp
FORCE_TYPE    PRESSURE      pres
FORCE_TYPE    SWDOWN        dswrf
FORCE_TYPE    LWDOWN        dlwrf
FORCE_TYPE    VP            vp
FORCE_TYPE    WIND          wind
WIND_H        10.0
```

```
PARAMETERS      /workspaces/VIC_sample_data/image/Stehekin/parameters/params.Stehekin.L2015.nc
LAI_SRC         FROM_VEGPARAM
FCAN_SRC        FROM_DEFAULT
ALB_SRC         FROM_VEGPARAM
NODES           3
SNOW_BAND       TRUE
```

```
RESULT_DIR   /workspaces/VIC_sample_data/sample_image
```

The first thing we need to check is

1) Whether the input file exists
2) Whether the output directory exists

Input file

Output directory

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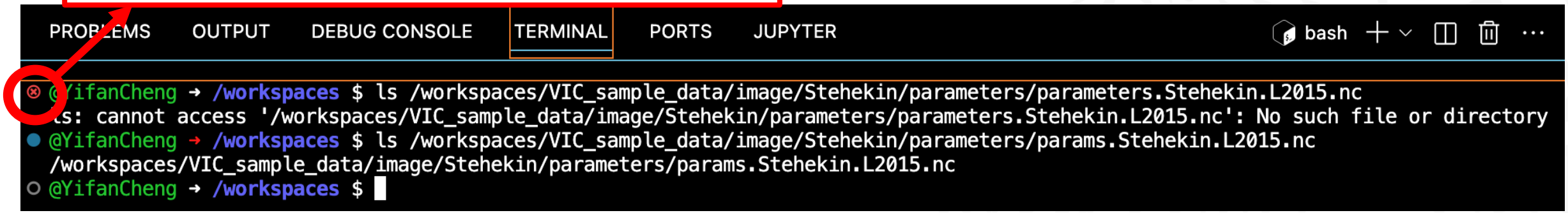
How can we check whether a file exists?

We can use `ls`

See example in next slides

How to check?

This red error indicates that this file does not exist!



If we find out that this file does not exist, it is important to check whether there is a **typo** in the file names or file paths.

Initial Conditions / State Files

- Initial conditions define the starting point of the simulation, including:
  - **Soil moisture/temperature**
  - **Snowpack**
  - **Groundwater levels**
  - **Streamflow**
  - **…**
- Initial conditions are crucial for ensuring the model starts from a realistic state, especially for short-term simulations or forecasting.

Initial Conditions / State Files

- VIC model output State Files that can be later used as an Initial Conditions file.

INIT_STATE: defines which initial condition file to use

```
#INIT_STATE   ${VIC_SAMPLE_DATA}/image/FindleyLake/state/state.FindleyLake.19740101_00000.nc
STATENAME     ${VIC_SAMPLE_DATA}/image/FindleyLake/state/state.FindleyLake
STATEYEAR     1980
STATEMONTH    1
STATEDAY      1
```

STATExxxx: defines the outputting of the date/time of the state as well as the state file name.

Cold Start and Spin-Up

- Most hydrologic models begin with a **cold start**, meaning they initialize with default or zeroed values for key state variables such as:
  - Soil moisture
  - Groundwater storage
  - Snowpack
  - Streamflow
- These initial values are often far from realistic, especially for long-term simulations or regions with complex hydrologic dynamics. As a result, the model needs a **spin-up period** to allow these variables to evolve and stabilize toward a more realistic equilibrium.

What does the configuration file look like for **spin-up runs**?

```
STARTYEAR     1970
STARTMONTH    1
STARTDAY      1
ENDYEAR       1973
ENDMONTH      12
ENDDAY        31
CALENDAR      PROLEPTIC_GREGORIAN
```

```
#INIT_STATE   ${VIC_SAMPLE_DATA}/image/FindleyLake/state/state.FindleyLake.19740101_00000.nc
STATENAME     ${VIC_SAMPLE_DATA}/image/FindleyLake/state/state.FindleyLake
STATEYEAR     1974
STATEMONTH    1
STATEDAY      1
```

1. We commented out the INIT_STATE, meaning that 1) we do not have an initial condition file, and 2) we start from cold start
2. We defined the STATENAME, STATEYEAR, STATEMONTH, and STATEDAY to output the state file, which will be used as an initial condition in the **production run**.

```
STARTYEAR     1974
STARTMONTH    1
STARTDAY      1
ENDYEAR       1979
ENDMONTH      12
ENDDAY        31
CALENDAR      PROLEPTIC_GREGORIAN
```

What does the configuration file look like for **production runs**?

```
INIT_STATE    ${VIC_SAMPLE_DATA}/image/FindleyLake/state/state.FindleyLake.19740101_00000.nc
STATENAME     ${VIC_SAMPLE_DATA}/image/FindleyLake/state/state.FindleyLake
STATEYEAR     1980
STATEMONTH    1
STATEDAY      1
```

1. We used the state file that was generated in the previous run as an initial condition file.
2. We usually save the state file for the last timestep in a model production run.

State versus ***Flux*** variables

- A **state variable** shows an instantaneous state at a specific time snapshot, or a mean state over a specific time step. It's an ***amount***, so its unit carries no time dimension (mm, m^3, °C).
  - Examples: soil moisture, snow water equivalent (SWE), groundwater storage, reservoir/lake storage, stream stage (water level).
- A **flux variable** shows — shows the ***rate*** at which water (or energy) crosses a boundary or moves between storages, over a time interval. Because it's a rate, its unit always has time in the denominator (mm/day, m^3/s, W/m^2).
  - Examples: precipitation, evapotranspiration, infiltration, runoff, streamflow discharge.

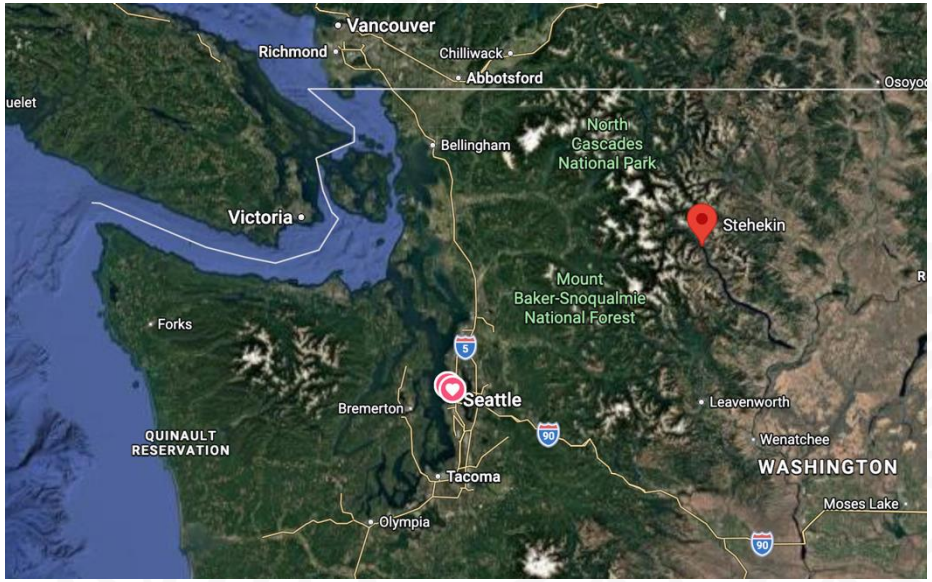
VIC output variables

- You might be curious why VIC output does not have the “time” dimension for its flux variables.
  - For example, 10 mm of precipitation means “10 mm**/timestep**” of precipitation

- List of VIC output variables
  - https://vic.readthedocs.io/en/master/Documentation/OutputVarList/

Can you check the water balance using the VIC model output?

- What is the Water Balance Equation?
  - $P = ET + R + \Delta S$
  - Note: Storage not only means soil moisture, but snowpack can be viewed as a temporary water storage



- If you were given a 10-day VIC model run for ***Stekehin, WA***, from Jan 1, 2026, to Jan 10, 2026. It has a total of 28 mm of precipitation; where did the precipitation go?
  - You were given OUT_EVAP, OUT_SOIL_MOIST, OUT_RUNOFF, OUT_BASEFLOW, OUT_SWE
  - What would you do?

Where did the precipitation go?

- Step 1: Identify which are state variables and which are flux variables
  - OUT_EVAP, OUT_RUNOFF, OUT_BASEFLOW, OUT_SOIL_MOIST, OUT_SWE

Variable	Type
OUT_EVAP	Flux
OUT_RUNOFF	Flux
OUT_BASEFLOW	Flux
OUT_SOIL_MOIST	State
OUT_SWE	State

Where did the precipitation go?

- Step 2: Identify the quantity of water for each variable
  - For state variables, we use the final state minus the beginning state
  - For flux variables, we aggregate all fluxes across all time steps

$$P \approx \Sigma EVAP + \Sigma RUNOFF + \Sigma BASEFLOW + \Delta SOIL_MOIST + \Delta SWE$$

Where did the precipitation go?

- **Three things to flag**
  - **OUT_SOIL_MOIST is usually per layer.** VIC typically reports soil moisture for each soil layer (2 or 3 layers) separately. We need to sum across layers to get total column storage
  - **What does a state variable's timestamp mean?** Does it represent an instantaneous state at that moment, or a mean state across a time step — and either way, is the timestamp placed at the beginning or the end of that step?
  - **Sign convention when you combine everything.** ΔS should be *added* to the flux terms with a consistent sign: if storage went up, that new water enters the system; if storage went down, some water leaves the system

References

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